EXERCISE

Tokens and projections

A new token, the same learned map

$x = (2, -1, 1)^T$, $W_Q = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & -1 \end{bmatrix}$. Find $q = W_Q x$.

A $(3, -3)^T$

B $(1, -1)^T$

C A three-coordinate vector

Vaswani et al. (2017), §3.2.1, Eq. 1 · Course example

SOLUTION

Tokens and projections

A new token, the same learned map

Apply the two output rows

x: input column; W_Q: 2 × 3 map; q: output column.

---

1. $q_1 = 1(2) + 0(-1) + 1(1) = 3.$
2. $q_2 = 0(2) + 2(-1) - 1(1) = -3.$
3. $q = (3, -3)^T$ has two coordinates.

Vaswani et al. (2017), §3.2.1, Eq. 1 · Course example

SOLUTION

Tokens and projections

A new token, the same learned map

Solution 2 / 2

Check the alternatives

q: projected token.

---

1. A is correct. B keeps only the last input feature’s contribution.
2. C keeps the input width, but the two weight rows define two outputs.

Vaswani et al. (2017), §3.2.1, Eq. 1 · Course example

RESULT

Tokens and projections

A new token, the same learned map

$x = (2, -1, 1)^T$, $W_Q = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & -1 \end{bmatrix}$. Find $q = W_Q x$.

- **A** $(3, -3)^T$ ✓
- **B** $(1, -1)^T$
- **C** A three-coordinate vector

Vaswani et al. (2017), §3.2.1, Eq. 1 · Course example

EXERCISE

Scaled dot-product attention

Alignment and length both affect the score

$q = (1, 0)^T$, $k_1 = (1, 0)^T$, $k_2 = (2, 0)^T$, $d_k = 2$. Which key has the larger score?

A Key 1

B Key 2

C They must tie

Vaswani et al. (2017), §3.2.1, Eq. 1 · Course example

SOLUTION

Scaled dot-product attention

Alignment and length both affect the score

Solution 1 / 1

Evaluate the dot products

q: query; k_1, k_2: keys; $d_k = 2$.

1. $q^T k_1 = 1$ and $q^T k_2 = 2$.
2. $s_1 = 1/\sqrt{2} < s_2 = 2/\sqrt{2}$.
3. B is correct. Equal directions alone do not imply equal dot products.

Vaswani et al. (2017), §3.2.1, Eq. 1 · Course example

RESULT

Scaled dot-product attention

Alignment and length both affect the score

$q = (1, 0)^T$, $k_1 = (1, 0)^T$, $k_2 = (2, 0)^T$, $d_k = 2$. Which key has the larger score?

A Key 1

B Key 2 ✓

C They must tie

Vaswani et al. (2017), §3.2.1, Eq. 1 · Course example

EXERCISE

Scaled dot-product attention

Unequal scores change the mixture

$s = (0, \log 3)$, $v_1 = (2, 0)^T$, $v_2 = (0, 4)^T$. Find the output.

A $(1, 2)^T$

B $(1/2, 3)^T$

C $(0, 4)^T$

Vaswani et al. (2017), §3.2.1, Eq. 1 · Course example

SOLUTION

Scaled dot-product attention

Unequal scores change the mixture

Solution 1 / 2

Convert scores to weights

s: score row; p: its softmax weights.

1. $e^{s} = (1, 3)$; the sum is 4.
2. $p = (1/4, 3/4)$.

Vaswani et al. (2017), §3.2.1, Eq. 1 · Course example

SOLUTION

Scaled dot-product attention

Unequal scores change the mixture

Solution 2 / 2

Mix the value vectors

$v_1 = (2, 0)^T$, $v_2 = (0, 4)^T$.

---

1. $y = \frac{1}{4} v_1 + \frac{3}{4} v_2 = (1/2, 3)^T$.
2. B is correct. A treats scores as equal; C replaces soft weighting with a hard maximum.

Vaswani et al. (2017), §3.2.1, Eq. 1 · Course example

RESULT

Scaled dot-product attention

Unequal scores change the mixture

$s = (0, \log 3)$, $v_1 = (2, 0)^T$, $v_2 = (0, 4)^T$. Find the output.

A $(1, 2)^T$

B $(1/2, 3)^T$ ✓

C $(0, 4)^T$

Vaswani et al. (2017), §3.2.1, Eq. 1 · Course example

EXERCISE

Scaled dot-product attention

The scale is determined by the assumptions

All coordinates of q, k are mutually independent, with mean 0 and variance 1.

For $u = q^T k$ with $d_k = 16$, find the unscaled standard deviation.

A 16

B 4

C 1

Vaswani et al. (2017), §3.2.1, Eq. 1 · Course example

SOLUTION

Scaled dot-product attention

The scale is determined by the assumptions

Solution 1 / 1

Separate variance from standard deviation

$u = q^T k$; $d_k = 16$.

1. $\mathrm{Var}(u) = 16$.
2. $\mathrm{SD}(u) = \sqrt{16} = 4$.
3. B is correct. A confuses variance with standard deviation; C describes the scaled score.

Vaswani et al. (2017), §3.2.1, Eq. 1 · Course example

RESULT

Scaled dot-product attention

The scale is determined by the assumptions

All coordinates of q, k are mutually independent, with mean 0 and variance 1.

For $u = q^T k$ with $d_k = 16$, find the unscaled standard deviation.

A 16

B 4 ✓

C 1

Vaswani et al. (2017), §3.2.1, Eq. 1 · Course example

EXERCISE

Stable attention and masks

A shared shift or a change of scale?

A score row is $s = (0, \log 3)$. Which change preserves its softmax weights?

- **A** Subtract 5 from both entries
- **B** Multiply both entries by 2
- **C** Add 5 only to the second entry

Blanchard, Higham & Higham (2021), §4 · Course example

SOLUTION

Stable attention and masks

A shared shift or a change of scale?

Solution 1 / 1

Identify the invariance

s: logits; softmax normalizes their exponentials.

---

1. $\mathrm{softmax}(s - 51) = \mathrm{softmax}(s)$.
2. $\mathrm{softmax}(2s) = (1/10, 9/10)$.
3. A is correct. B changes the score gap; C multiplies only the second exponential by e^5.

Blanchard, Higham & Higham (2021), §4 · Course example

RESULT

Stable attention and masks

A shared shift or a change of scale?

A score row is $s = (0, \log 3)$. Which change preserves its softmax weights?

- **A** Subtract 5 from both entries ✓
- **B** Multiply both entries by 2
- **C** Add 5 only to the second entry

Blanchard, Higham & Higham (2021), §4 · Course example

EXERCISE

Stable attention and masks

Mask before normalizing

Scores are $(0, \log 3)$; the second key is forbidden. Which output weights are correct?

A $(1, 0)$

B $(1/4, 0)$

C $(1/4, 3/4)$

Vaswani et al. (2017), §3.2.1, Eq. 1 · Course example

SOLUTION

Stable attention and masks

Mask before normalizing

Solution 1 / 2

Normalize only over permitted keys

s: score row; key 2 is unavailable.

---

1. $s + M = (0, -\infty)$.
2. $\exp(s + M) = (1, 0)$ and its sum is 1.
3. $p = (1, 0)$. A is correct.

Vaswani et al. (2017), §3.2.1, Eq. 1 · Course example

SOLUTION

Stable attention and masks

Mask before normalizing

Solution 2 / 2

Why the other operations differ

p: normalized attention weights.

1. B zeroes a probability after normalization and leaves row sum 1/4.
2. C ignores the mask. Renormalizing allowed probabilities could work, but masking logits is direct.

Vaswani et al. (2017), §3.2.1, Eq. 1 · Course example

RESULT

Stable attention and masks

Mask before normalizing

Scores are $(0, \log 3)$; the second key is forbidden. Which output weights are correct?

- **A** $(1, 0)$ ✓
- **B** $(1/4, 0)$
- **C** $(1/4, 3/4)$

Vaswani et al. (2017), §3.2.1, Eq. 1 · Course example

EXERCISE

Stable attention and masks

Diagnose the NaN without changing the task

A padded query has no allowed key. Stable softmax returns NaN (Not a Number). Which repair preserves the declared information rule?

A. Use uniform weights over all keys

B. Mark the query invalid and output zero

C. Remove all masks for the batch

Vaswani et al. (2017), §3.2.1, Eq. 1 · Course example

SOLUTION

Stable attention and masks

Diagnose the NaN without changing the task

Solution 1 / 1

Separate invalid queries from valid predictions

M: mask; padded queries are excluded from the loss.

1. Detect a row with no permitted key before normalization.
2. For that invalid row, use zero weights/output and exclude its loss.
3. B is correct. A and C read forbidden entries.

Vaswani et al. (2017), §3.2.1, Eq. 1 · Course example

RESULT

Stable attention and masks

Diagnose the NaN without changing the task

A padded query has no allowed key. Stable softmax returns NaN (Not a Number). Which repair preserves the declared information rule?

A. Use uniform weights over all keys

B. Mark the query invalid and output zero ✓

C. Remove all masks for the batch

Vaswani et al. (2017), §3.2.1, Eq. 1 · Course example

EXERCISE

Stable attention and masks

Transfer: change the permitted information

$d_k = 2$, $q = (\sqrt{2}, 0)^T$, $k_1 = (0, 1)^T$, $k_2 = (\log 3, 0)^T$; $v_1 = (2, 0)^T$, $v_2 = (0, 4)^T$. Only key 1 is allowed. Find the output.

A $(1/2, 3)^T$

B $(2, 0)^T$

C $(0, 0)^T$

Vaswani et al. (2017), §3.2.1, Eq. 1 · Course example

SOLUTION

Stable attention and masks

Transfer: change the permitted information

Solution 1 / 1

Use the same scores with a different mask

s: scaled scores; p: allowed-key weights.

1. $s = (0, \log 3)$, but $s + M = (0, -\infty)$.
2. $p = (1, 0)$, so $y = 1v_1 + 0v_2 = (2, 0)^T$.
3. B is correct. A ignores masking; C discards a valid key.

Vaswani et al. (2017), §3.2.1, Eq. 1 · Course example

RESULT

Stable attention and masks

Transfer: change the permitted information

$d_k = 2$, $q = (\sqrt{2}, 0)^T$, $k_1 = (0, 1)^T$, $k_2 = (\log 3, 0)^T$; $v_1 = (2, 0)^T$, $v_2 = (0, 4)^T$. Only key 1 is allowed. Find the output.

A $(1/2, 3)^T$

B $(2, 0)^T$ ✓

C $(0, 0)^T$

Vaswani et al. (2017), §3.2.1, Eq. 1 · Course example

EXERCISE

Multi-head attention

Trace a tensor through the heads

For self-attention, $B = 2$, $T = 3$, $D = 8$, $H = 2$. Find the query shape Q after splitting and transposing, and the score shape S.

A Q [2,2,3,4]; S [2,2,3,3]

B Q [2,3,2,4]; S [2,3,2,2]

C Q [2,2,3,8]; S [2,2,8,8]

Vaswani et al. (2017), §3.2.2 · Course example

SOLUTION

Multi-head attention

Trace a tensor through the heads

Name each axis

B: batch; T: tokens; H: heads; $d_h = D/H$.

1. $d_h = 8/2 = 4$.
2. $Q : [B, H, T, d_h] = [2, 2, 3, 4]$.
3. $S : [B, H, T, T] = [2, 2, 3, 3]$.

Vaswani et al. (2017), §3.2.2 · Course example

SOLUTION

Multi-head attention

Trace a tensor through the heads

Solution 2 / 2

Check the alternatives

S: query-by-key scores in each head.

1. A is correct. B omits the token/head transpose.
2. C fails to split the width and confuses features with token counts.

Vaswani et al. (2017), §3.2.2 · Course example

RESULT

Multi-head attention

Trace a tensor through the heads

For self-attention, $B = 2$, $T = 3$, $D = 8$, $H = 2$. Find the query shape Q after splitting and transposing, and the score shape S.

- **A** Q [2,2,3,4]; S [2,2,3,3] ✓
- **B** Q [2,3,2,4]; S [2,3,2,2]
- **C** Q [2,2,3,8]; S [2,2,8,8]

Vaswani et al. (2017), §3.2.2 · Course example

EXERCISE

Multi-head attention

Transfer: a different output map

The joined head vector is $c = (1, 2, 3, -1)^T$. The output map is $W_O = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & -1 \end{bmatrix}$. Find $W_O c$.

A $(3, 2)^T$

B $(4, 1)^T$

C $(1, 2, 3, -1)^T$

Vaswani et al. (2017), §3.2.2 · Course example

SOLUTION

Multi-head attention

Transfer: a different output map

Solution 1 / 1

Read the new map

c: concatenated heads; W_O: two output rows.

1. First output: $0 + 0 + 3 + 0 = 3$.
2. Second output: $1 + 0 + 0 - (-1) = 2$.
3. A is correct. B reuses the earlier map; C stops before projection.

Vaswani et al. (2017), §3.2.2 · Course example

RESULT

Multi-head attention

Transfer: a different output map

The joined head vector is $c = (1, 2, 3, -1)^T$. The output map is $W_O = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & -1 \end{bmatrix}$. Find $W_O c$.

A $(3, 2)^T$ ✓

B $(4, 1)^T$

C $(1, 2, 3, -1)^T$

Vaswani et al. (2017), §3.2.2 · Course example

EXERCISE

Multi-head attention

Correct shapes can hide a wrong normalization

Scores have shape [B,H,Tq,Tk]: batch, heads, queries, keys. A programmer uses `softmax(S,dim=-2)`. What is normalized?

A. Keys for each query

B. Queries for each key

C. Features of each token

Vaswani et al. (2017), §3.2.2 · Course example

SOLUTION

Multi-head attention

Correct shapes can hide a wrong normalization

Solution 1 / 1

Read the axis labels

$S_{b,h,i,j}$ compares query i with key j.

1. The second-last axis is i, the query index.
2. dim=−2 makes columns sum to one across queries.
3. B describes the bug. Correct attention uses dim=−1 for key normalization.

Vaswani et al. (2017), §3.2.2 · Course example

RESULT

Multi-head attention

Correct shapes can hide a wrong normalization

Scores have shape [B,H,Tq,Tk]: batch, heads, queries, keys. A programmer uses `softmax(S,dim=-2)`. What is normalized?

A Keys for each query

B Queries for each key ✓

C Features of each token

Vaswani et al. (2017), §3.2.2 · Course example

EXERCISE

Multi-head attention

Transfer: broadcast a key-validity mask

Scores have shape [2,3,4,5]: batch, heads, queries, keys. Key validity has shape [2,5], with True for allowed keys. Which reshape shares this key rule across every head and query?

A [2,1,1,5]

B [2,5,1,1]

C [1,1,2,5]

Vaswani et al. (2017), §3.2.2 · Course example

SOLUTION

Multi-head attention

Transfer: broadcast a key-validity mask

Solution 1 / 1

Align the batch and key axes

$B = 2, H = 3, T_q = 4, T_k = 5.$

1. Use singleton head and query axes: [2,1,1,5].
2. Broadcast those axes to [2,3,4,5]; A is correct.
3. B cannot broadcast: head extent 5 differs from 3. C cannot broadcast: query extent 2 differs from 4.

Vaswani et al. (2017), §3.2.2 · Course example

RESULT

Multi-head attention

Transfer: broadcast a key-validity mask

Scores have shape [2,3,4,5]: batch, heads, queries, keys. Key validity has shape [2,5], with True for allowed keys. Which reshape shares this key rule across every head and query?

- A [2,1,1,5] ✓
- B [2,5,1,1]
- C [1,1,2,5]

Vaswani et al. (2017), §3.2.2 · Course example

EXERCISE

Position and order

What does content alone fail to identify?

The same two feature vectors are presented in opposite orders to unmasked attention with no position features.

A. Dot products recover their original order

B. Outputs move with the input permutation

C. Every output becomes zero

Vaswani et al. (2017), §3.5 · Course example

SOLUTION

Position and order

What does content alone fail to identify?

Solution 1 / 1

Apply the stated symmetry

X, Y: input/output token rows; Π: swap of the two rows.

1. $X \mapsto \Pi X$ implies $Y \mapsto \Pi Y$.
2. B is correct: relationships are preserved without an original-position label.
3. A invents missing information; C does not follow from equivariance.

Vaswani et al. (2017), §3.5 · Course example

RESULT

Position and order

What does content alone fail to identify?

The same two feature vectors are presented in opposite orders to unmasked attention with no position features.

A Dot products recover their original order

B Outputs move with the input permutation ✓

C Every output becomes zero

Vaswani et al. (2017), §3.5 · Course example

EXERCISE

Position and order

Transfer: one sinusoidal pair at a new position

One position pair is $(\sin(p\omega), \cos(p\omega))$. For this illustrative frequency $\omega = \pi/2$, move from position p=0 to p=1.

A (1,0)

B (0,1)

C (−1,0)

Vaswani et al. (2017), §3.5 · Course example

SOLUTION

Position and order

Transfer: one sinusoidal pair at a new position

Solution 1 / 1

Evaluate in the stated coordinate order

$\omega = \pi/2$; the pair is sine first, cosine second.

---

1. At $p = 0$: $(\sin 0, \cos 0) = (0, 1)$.
2. At $p = 1$: $(\sin(\pi/2), \cos(\pi/2)) = (1, 0)$.
3. A is correct. B keeps the old position; C uses the wrong sine sign.

Vaswani et al. (2017), §3.5 · Course example

RESULT

Position and order

Transfer: one sinusoidal pair at a new position

One position pair is $(\sin(p\omega), \cos(p\omega))$. For this illustrative frequency $\omega = \pi/2$, move from position p=0 to p=1.

- **A** (1,0) ✓
- **B** (0,1)
- **C** (−1,0)

Vaswani et al. (2017), §3.5 · Course example

EXERCISE

Position and order

Transfer: shift both positions

Unrotated pairs: $q = k = (1, 0)^T$; frequency $\omega = \pi/2$. Change query/key positions (p, s) from $(0, 1)$ to $(2, 3)$. Find the rotated query–key dot product before attention scaling.

A 0

B 1

C -1

Su et al., RoFormer (2021), §3.2 · Course example

SOLUTION

Position and order

Transfer: shift both positions

Solution 1 / 1

Use the relative angle

p, s: query/key positions; content unchanged.

---

1. $(s-p)\omega = (3-2)\pi/2 = \pi/2.$
2. $q^T R(\pi/2)k = 0.$
3. A is correct. B uses zero offset; C uses a half-turn.

Su et al., RoFormer (2021), §3.2 · Course example

RESULT

Position and order

Transfer: shift both positions

Unrotated pairs: $q = k = (1, 0)^T$; frequency $\omega = \pi/2$. Change query/key positions (p, s) from $(0, 1)$ to $(2, 3)$. Find the rotated query–key dot product before attention scaling.

A 0 ✓

B 1

C -1

Su et al., RoFormer (2021), §3.2 · Course example

EXERCISE

Transformer blocks

Transfer: normalize one token

One token is $x = (1, 3)$. Use $\gamma = (1, 1)$, $\beta = (0, 0)$ and $\epsilon = 0$ for this nonconstant numerical example.

A (−1,1)

B (−1/√2, 1/√2)

C (1,3)

Xiong et al. (2020), §3.1, Table 1 · Course example

SOLUTION

Transformer blocks

Transfer: normalize one token

Solution 1 / 2

Use the feature mean and mean squared deviation

$D = 2$ features; no batch or token averaging.

---

1. $\mu = (1 + 3)/2 = 2$; $x - \mu = (-1, 1)$.
2. $v = [(-1)^2 + 1^2]/2 = 1$.
3. $\hat{x} = (-1, 1)/\sqrt{1} = (-1, 1)$. A is correct.

Xiong et al. (2020), §3.1, Table 1 · Course example

SOLUTION

Transformer blocks

Transfer: normalize one token

Solution 2 / 2

Diagnose the alternative statistics

$\gamma = 1$ and $\beta = 0$ preserve the normalized values.

1. B uses the sample-variance divisor D−1; LayerNorm uses D.
2. C omits centering and scaling. In real implementations keep positive epsilon.

Xiong et al. (2020), §3.1, Table 1 · Course example

RESULT

Transformer blocks

Transfer: normalize one token

One token is $x = (1, 3)$. Use $\gamma = (1, 1)$, $\beta = (0, 0)$ and $\epsilon = 0$ for this nonconstant numerical example.

- **A** $(-1,1)$ ✓
- **B** $(-1/\sqrt{2}, 1/\sqrt{2})$
- **C** $(1,3)$

Xiong et al. (2020), §3.1, Table 1 · Course example

EXERCISE

Transformer blocks

A residual connection constrains the shape

The residual input X has shape [2,5,12]: batch, tokens, features. Its attention branch returns [2,5,8]. Can they be directly added?

A. Yes, ordinary broadcasting repairs it

B. No, the branch must return width 12

C. Yes, because both have five tokens

Xiong et al. (2020), §3.1, Table 1 · Course example

SOLUTION

Transformer blocks

A residual connection constrains the shape

Solution 1 / 1

Compare corresponding coordinates

X: residual input; branch: learned update.

1. Feature widths 12 and 8 differ and neither is one.
2. Ordinary broadcasting cannot align these feature coordinates.
3. B is correct: use the output projection to return width 12.

Xiong et al. (2020), §3.1, Table 1 · Course example

RESULT

Transformer blocks

A residual connection constrains the shape

The residual input X has shape [2,5,12]: batch, tokens, features. Its attention branch returns [2,5,8]. Can they be directly added?

A Yes, ordinary broadcasting repairs it

B No, the branch must return width 12 ✓

C Yes, because both have five tokens

Xiong et al. (2020), §3.1, Table 1 · Course example

EXERCISE

Transformer blocks

Transfer: a missing path changes the architecture

$Y = X + FFN(MHA(X))$ is described as the pre-LN block. Here MHA is multi-head attention and FFN is the tokenwise feed-forward network. What is missing?

A. Only position information

B. Both normalizations and the intermediate residual

C. A final softmax only

Xiong et al. (2020), §3.1, Table 1 · Course example

SOLUTION

Transformer blocks

Transfer: a missing path changes the architecture

Solution 1 / 1

Compare the computation graph

U: intermediate residual representation.

1. $U = X + MHA(LN(X))$.
2. $Y = U + FFN(LN(U))$.
3. B is correct. Position features or final softmax do not restore these missing paths.

Xiong et al. (2020), §3.1, Table 1 · Course example

RESULT

Transformer blocks

Transfer: a missing path changes the architecture

$Y = X + FFN(MHA(X))$ is described as the pre-LN block. Here MHA is multi-head attention and FFN is the tokenwise feed-forward network. What is missing?

A. Only position information

B. Both normalizations and the intermediate residual ✓

C. A final softmax only

Xiong et al. (2020), §3.1, Table 1 · Course example

EXERCISE

Cost and the KV cache

Transfer: a longer sequence

The materialized score tensor has shape [B,H,T,T]: batch, heads, queries, keys.

Keep B, H and bytes per entry fixed. Double T. How does its memory change?

A ×2

B ×4

C Unchanged

Vaswani et al. (2017), §3.2.2 · Course example

SOLUTION

Cost and the KV cache

Transfer: a longer sequence

Solution 1 / 1

Count both token axes

S has shape [B,H,T,T].

---

1. $BH(2T)^2 = 4BHT^2$.
2. B is correct. A counts only the extra queries and forgets the extra keys.
3. C confuses fixed feature width with a fixed pair count.

Vaswani et al. (2017), §3.2.2 · Course example

RESULT

Cost and the KV cache

Transfer: a longer sequence

The materialized score tensor has shape [B,H,T,T]: batch, heads, queries, keys.

Keep B, H and bytes per entry fixed. Double T. How does its memory change?

A ×2

B ×4 ✓

C Unchanged

Vaswani et al. (2017), §3.2.2 · Course example

EXERCISE

Cost and the KV cache

Transfer: can this memory be reused?

Camera tokens are replaced by a new frame. Can their old cross-attention keys and values serve as exact representations of the new frame?

A. Yes, when shapes match

B. No, recompute the changed memory

C. Yes, if action horizon is fixed

Shazeer (2019), §2.4–2.4.1 · Course example

SOLUTION

Cost and the KV cache

Transfer: can this memory be reused?

Solution 1 / 1

Trace projections back to their inputs

E: encoded camera tokens; K,V are their projections.

---

1. $K = EW_K^T, V = EW_V^T$.
2. A new frame can change E and both projections.
3. B is correct. Matching shape or horizon does not imply matching values.

Shazeer (2019), §2.4–2.4.1 · Course example

RESULT

Cost and the KV cache

Transfer: can this memory be reused?

Camera tokens are replaced by a new frame. Can their old cross-attention keys and values serve as exact representations of the new frame?

A Yes, when shapes match

B No, recompute the changed memory ✓

C Yes, if action horizon is fixed

Shazeer (2019), §2.4–2.4.1 · Course example

EXERCISE

Cost and the KV cache

Transfer: count incremental comparisons

Generate four tokens from an empty prefix, one head. Count dense query–key dot products before masking, including each current token. Compare cached decoding with a full t × t score matrix recomputed at every step.

A Cached 10; recomputed 30

B Cached 4; recomputed 16

C Cached 16; recomputed 16

Shazeer (2019), §2.4–2.4.1 · Course example

SOLUTION

Cost and the KV cache

Transfer: count incremental comparisons

Solution 1 / 1

Count dense scores over four decoding steps

$t = 1, 2, 3, 4$; count dot products before causal masking.

---

1. Cached: $1 + 2 + 3 + 4 = 10$.
2. Full-prefix dense reruns: $1^2 + 2^2 + 3^2 + 4^2 = 30$.
3. A is correct. B drops past keys and repeated passes; C counts only one final full-sequence pass.

Shazeer (2019), §2.4–2.4.1 · Course example

RESULT

Cost and the KV cache

Transfer: count incremental comparisons

Generate four tokens from an empty prefix, one head. Count dense query–key dot products before masking, including each current token. Compare cached decoding with a full t × t score matrix recomputed at every step.

- A Cached 10; recomputed 30 ✓
- B Cached 4; recomputed 16
- C Cached 16; recomputed 16

Shazeer (2019), §2.4–2.4.1 · Course example

EXERCISE

Action chunking

Chunk length is not execution length

At prediction time t, a policy predicts K=6 actions and executes the first E=2 before observing again. K is chunk length; E is execution-prefix length. Which target times from this chunk are executed?

A Times t through t+5

B Times t and t+1

C Only time t+5

Zhao et al. (2023), §IV-A · Course example

SOLUTION

Action chunking

Chunk length is not execution length

Solution 1 / 1

Separate prediction and execution

$K = 6$: predicted rows; $E = 2$: executed prefix length.

1. The chunk covers target times t,…,t+5.
2. Its first two rows target t and t+1.
3. B is correct. A confuses K with E; C chooses the last row rather than the prefix.

Zhao et al. (2023), §IV-A · Course example

RESULT

Action chunking

Chunk length is not execution length

At prediction time t, a policy predicts K=6 actions and executes the first E=2 before observing again. K is chunk length; E is execution-prefix length. Which target times from this chunk are executed?

A Times t through t+5

B Times t and t+1 ✓

C Only time t+5

Zhao et al. (2023), §IV-A · Course example

EXERCISE

Temporal ensembling

Transfer: three estimates for one target

Oldest-to-newest scalar predictions for the same target s are $1, 3, 5$. Use $w_j = e^{-mj}$ with $j = 0, 1, 2$ and $m = \log 2$. Find the normalized ensemble.

A 3

B 15/7

C 27/7

Zhao et al. (2023), §IV-A · Course example

SOLUTION

Temporal ensembling

Transfer: three estimates for one target

Solution 1 / 2

Weight, sum, normalize

$j = 0, 1, 2$: oldest-to-newest; all predictions target s.

1. $w = (1, 1/2, 1/4)$, with sum $7/4$.
2. Weighted sum $= 1 + 3/2 + 5/4 = 15/4$.
3. $\bar{a}_s = (15/4)/(7/4) = 15/7$.

Zhao et al. (2023), §IV-A · Course example

SOLUTION

Temporal ensembling

Transfer: three estimates for one target

Solution 2 / 2

Interpret the result

$\bar{a}_s$: ensemble action.

---

1. B is correct. A uses equal weights.
2. C reverses the weights and favors the newest prediction.

Zhao et al. (2023), §IV-A · Course example

RESULT

Temporal ensembling

Transfer: three estimates for one target

Oldest-to-newest scalar predictions for the same target s are $1, 3, 5$. Use $w_j = e^{-mj}$ with $j = 0, 1, 2$ and $m = \log 2$. Find the normalized ensemble.

A 3

B 15/7 ✓

C 27/7

Zhao et al. (2023), §IV-A · Course example

EXERCISE

ACT: observations to actions

Transfer: encoder tokens and action slots

A model has 20 observation tokens and predicts a chunk of 6 actions. Each action query reads every observation token. What is one head’s cross-attention score shape?

A. 20×20

B. 6×20

C. 6×6

Zhao et al. (2023), §IV-B–C, Appendix C · Course example

SOLUTION

ACT: observations to actions

Transfer: encoder tokens and action slots

Solution 1 / 1

Identify query and key collections

$K = 6$: action queries; $T_o = 20$: observation keys.

1. $Q_a : [6, d_k]$, $K_o^T : [d_k, 20]$.
2. $Q_a K_o^T : [6, 20]$.
3. B is correct. A and C replace cross-attention with self-attention on one collection.

Zhao et al. (2023), §IV-B–C, Appendix C · Course example

RESULT

ACT: observations to actions

Transfer: encoder tokens and action slots

A model has 20 observation tokens and predicts a chunk of 6 actions. Each action query reads every observation token. What is one head’s cross-attention score shape?

A. 20×20

B. 6×20 ✓

C. 6×6

Zhao et al. (2023), §IV-B–C, Appendix C · Course example

EXERCISE

ACT: latent-variable training

Log-variance is not the standard deviation

One latent coordinate has $\mu = 1$, $\ell = \log \sigma^2 = \log 4$, and $\epsilon = -1$. Use $z = \mu + \sigma\epsilon$. Find z.

A -3

B -1

C 1

Kingma & Welling (2014), §2, Appendix B · Course example

SOLUTION

ACT: latent-variable training

Log-variance is not the standard deviation

Solution 1 / 1

Convert the parameter before sampling

ℓ: log-variance; σ: standard deviation.

1. $\sigma^2 = e^{\ell} = 4$, so $\sigma = e^{\ell/2} = 2$.
2. $z = \mu + \sigma\epsilon = 1 + 2(-1) = -1$.
3. B is correct. A uses variance as scale; C ignores the sampled noise.

Kingma & Welling (2014), §2, Appendix B · Course example

RESULT

ACT: latent-variable training

Log-variance is not the standard deviation

One latent coordinate has $\mu = 1$, $\ell = \log \sigma^2 = \log 4$, and $\epsilon = -1$. Use $z = \mu + \sigma\epsilon$. Find z.

A -3

B -1 ✓

C 1

Kingma & Welling (2014), §2, Appendix B · Course example

EXERCISE

ACT: latent-variable training

Transfer: which divergence creates the gap?

For a fixed chunk A and observation o, let $q(z)$ be the encoder density. The exact identity is $\log p_\theta(A \mid o) = \mathcal{L}_{ELBO} + G$. Which G is the gap?

A $D_{KL}(q \| p_\theta(z \mid A, o))$

B $-D_{KL}(q \| p(z))$

C The action prediction error alone

Kingma & Welling (2014), §2, Appendix B · Course example

SOLUTION

ACT: latent-variable training

Transfer: which divergence creates the gap?

Solution 1 / 1

Distinguish two different distributions

q: encoder; posterior: conditioned on the observed target.

---

1. $G = D_{KL}(q \| \text{true posterior}) \geq 0$.
2. The prior KL is already inside $\mathcal{L}_{ELBO}$ with a minus sign.
3. A is correct. B confuses regularization with the posterior approximation gap.

Kingma & Welling (2014), §2, Appendix B · Course example

RESULT

ACT: latent-variable training

Transfer: which divergence creates the gap?

For a fixed chunk A and observation o, let $q(z)$ be the encoder density. The exact identity is $\log p_\theta(A \mid o) = \mathcal{L}_{ELBO} + G$. Which G is the gap?

A $D_{KL}(q \| p_\theta(z \mid A, o))$ ✓

B $-D_{KL}(q \| p(z))$

C The action prediction error alone

Kingma & Welling (2014), §2, Appendix B · Course example

EXERCISE

ACT: latent-variable training

Transfer: combine the two loss terms

Four valid action-coordinate errors are $1, -1, 0, 2$. A scalar Gaussian encoder has $\mu = 1$, $\sigma^2 = 1$; its prior is $\mathcal{N}(0, 1)$. Use mean L1 and $\beta = 2$. Find $\mathcal{L}_{train}$.

A 1

B 2

C 5

Zhao et al. (2023), §IV-B–C, Appendix C · Course example

SOLUTION

ACT: latent-variable training

Transfer: combine the two loss terms

Reconstruct with the stated reduction

$Kd_a = 4$ action coordinates; errors are prediction minus target.

1. $\mathcal{L}_{rec} = (|1| + |-1| + |0| + |2|)/4 = 1.$
2. $D_{KL} = \frac{1}{2}(1 + 1 - 1 - \log 1) = 1/2.$

Zhao et al. (2023), §IV-B–C, Appendix C · Course example

SOLUTION

ACT: latent-variable training

Transfer: combine the two loss terms

Solution 2 / 2

Combine without changing the reduction

$\beta = 2$; mean reconstruction error equals 1.

---

1. $\mathcal{L}_{train} = 1 + 2(1/2) = 2.$
2. B is correct. A drops the KL term; C uses summed rather than mean L1.

Zhao et al. (2023), §IV-B–C, Appendix C · Course example

RESULT

ACT: latent-variable training

Transfer: combine the two loss terms

Four valid action-coordinate errors are $1, -1, 0, 2$. A scalar Gaussian encoder has $\mu = 1$, $\sigma^2 = 1$; its prior is $\mathcal{N}(0, 1)$. Use mean L1 and $\beta = 2$. Find $\mathcal{L}_{train}$.

A 1

B 2 ✓

C 5

Zhao et al. (2023), §IV-B–C, Appendix C · Course example

EXERCISE

ACT: deployment

Training and inference use different information

The robot receives current images and joint measurements. Which input route matches the reported ACT inference procedure?

A. Encode the unknown future target chunk

B. Current observations plus z=0

C. Draw z from the demonstration posterior

Zhao et al. (2023), §IV-B–C, Appendix C · Course example

SOLUTION

ACT: deployment

Training and inference use different information

Solution 1 / 1

Identify what is available now

o_t: current observations; the future demonstration is unavailable.

---

1. Discard the demonstration-conditioned latent encoder.
2. Set z to the standard-prior mean, zero, and run the decoder.
3. B is correct. A and C require future target information.

Zhao et al. (2023), §IV-B–C, Appendix C · Course example

RESULT

ACT: deployment

Training and inference use different information

The robot receives current images and joint measurements. Which input route matches the reported ACT inference procedure?

A Encode the unknown future target chunk

B Current observations plus z=0 ✓

C Draw z from the demonstration posterior

Zhao et al. (2023), §IV-B–C, Appendix C · Course example

EXERCISE

ACT: deployment

Transfer: trace the live policy loop

At target time s, two chunks predict scalar actions 2 and 4, oldest first. Use $w_j = e^{-mj}$ with $m = \log 2$. A fresh observation arrives after execution. What target is executed, and what computation follows?

A. Execute 8/3; encode the fresh observation

B. Execute 3; change β

C. Execute 4; reuse all old image features

Zhao et al. (2023), §IV-B–C, Appendix C · Course example

SOLUTION

ACT: deployment

Transfer: trace the live policy loop

Solution 1 / 2

Compute the same-time ensemble

$r_0 < r_1$; both predictions target s.

1. $w = (1, 1/2)$; normalize to $(2/3, 1/3)$.
2. $\bar{a}_s = (2/3)2 + (1/3)4 = 8/3$.

Zhao et al. (2023), §IV-B–C, Appendix C · Course example

SOLUTION

ACT: deployment

Transfer: trace the live policy loop

Solution 2 / 2

Close the feedback loop

o_{s+1}: fresh observations; inference latent remains zero.

1. Encode o_{s+1} and predict $\widehat{A}_{s+1} = f_\theta(o_{s+1}, 0)$.
2. A is correct. B changes the ensemble and a training coefficient. C discards the stated weighting and fresh evidence.

Zhao et al. (2023), §IV-B–C, Appendix C · Course example

RESULT

ACT: deployment

Transfer: trace the live policy loop

At target time s, two chunks predict scalar actions 2 and 4, oldest first. Use $w_j = e^{-mj}$ with $m = \log 2$. A fresh observation arrives after execution. What target is executed, and what computation follows?

- **A** Execute 8/3; encode the fresh observation ✓
- **B** Execute 3; change β
- **C** Execute 4; reuse all old image features

Zhao et al. (2023), §IV-B–C, Appendix C · Course example

EXERCISE

ACT: deployment

Transfer: a changed scene after prediction

A chunk predicts six actions. After one executed action, the object moves.

Which response uses the new observation?

A. Keep executing all old targets without recomputing

B. Encode the new observation and predict a new chunk

C. Change the KL coefficient at inference

Zhao et al. (2023), §IV-B–C, Appendix C · Course example

SOLUTION

ACT: deployment

Transfer: a changed scene after prediction

Solution 1 / 1

Locate the input that changed

o: observation; learned model parameters remain fixed.

1. Object motion changes the current observation.
2. Re-encoding it can change attention memory and future predicted actions.
3. B uses that new evidence. A is open-loop; C changes a training parameter rather than reading the scene.

Zhao et al. (2023), §IV-B–C, Appendix C · Course example

RESULT

ACT: deployment

Transfer: a changed scene after prediction

A chunk predicts six actions. After one executed action, the object moves.

Which response uses the new observation?

A. Keep executing all old targets without recomputing

B. Encode the new observation and predict a new chunk ✓

C. Change the KL coefficient at inference

Zhao et al. (2023), §IV-B–C, Appendix C · Course example

EXERCISE

Discrete tokens and continuous actions

Transfer: the retained set depends on mass

Sorted token probabilities are $(0.5, 0.3, 0.2)$ and the top-p threshold is $\rho = 0.7$. Retain the smallest prefix reaching ρ, then renormalize. Which distribution is sampled?

A $(0.5, 0.3, 0.2)$

B $(0.625, 0.375, 0)$

C $(1, 0, 0)$

Holtzman et al. (2020), §3.1 · Course example

SOLUTION

Discrete tokens and continuous actions

Transfer: the retained set depends on mass

Solution 1 / 1

Find the smallest sufficient prefix

$\rho = 0.7$: cumulative-mass threshold.

---

1. $0.5 < 0.7$, while $0.5 + 0.3 = 0.8 \geq 0.7$.
2. Retain two tokens and divide by 0.8: $(5/8, 3/8, 0)$.
3. B is correct. A omits truncation; C stops before reaching the threshold.

Holtzman et al. (2020), §3.1 · Course example

RESULT

Discrete tokens and continuous actions

Transfer: the retained set depends on mass

Sorted token probabilities are $(0.5, 0.3, 0.2)$ and the top-p threshold is $\rho = 0.7$. Retain the smallest prefix reaching ρ, then renormalize. Which distribution is sampled?

A $(0.5, 0.3, 0.2)$

B $(0.625, 0.375, 0)$ ✓

C $(1, 0, 0)$

Holtzman et al. (2020), §3.1 · Course example